T_1 – Grass (Producer) = 30,000 J of energy

T_2 – Rabbit (Primary Consumer) = ?

T_3 – Snake (Secondary Consumer) = ?

According to the ten percent law, during the transfer of energy, only about 10% of the energy flows from each trophic level to the next lower trophic level. So 10% of energy from T_1 gets transferred to T_2

So T_2 - Rabbit (primary consumer) receives $30000 \text{ X } \frac{10}{100} = 3000 \text{ J}$

Similarly, 10% of energy from T_2 gets transferred to T_3

So T_3 – Snake (Secondary consumer) receives $3000 \text{ X } \frac{10}{100} = 300 \text{ J}$

Answer:

1. The third tropic level T_3 – (Snake) receives 300 J of energy.

Exercise 16: Determination of Population density and Percentage frequency by Quadrat method.

NOTE: Teachers can take the students to open space and teach them how to construct plot/ quadrats and to record the number of individuals of each plant species occurring in the quadrat. The percentage frequency should be calculated and entered in the practical observation and record note book. Examiner need not consider this experiment for Board Practical Examinations.

Aim:

To study population density and percentage frequency of different plant species of a given area by quadrat method.

Principle:

The number of individuals in a population never remains constant. It may increase or decrease due to many factors like birth rate, death rate, migration, etc. The number of individuals of a species presents per unit area or space of a given time is called population density. The population density and percentage frequency of different plant species can be determined by laying quadrats / segments of suitable size and recording of the number of individuals of each species occurring in the quadrat.

Requirements:

Metre scale, string or cord, hammer, nails, paper, pencil, etc.

Procedure:

1. In the selected site of study, hammer the nails firmly in the soil without damaging the vegetation.
2. Fix four nails to make a square plot.
3. Tie each end of the nails using a thread, to make 1 m X 1 m plot.
4. If the number of plants in the plot is large, the plot can be divided into quadrats.
5. Count the number of individuals of a species "A" present in the first quadrat and record the data in the table.
6. Similarly count the individuals of the species "A" in other quadrats respectively and record the data in the table.

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